

An odd-order weighing matrix refutes Conjecture 00000006334

Local solution draft
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Verdict: false. There is a weighing matrix of order 7 and weight 4. Its odd order directly contradicts the even-order clause of the conjecture.

Statement and standard definition. The English source explicitly asserts that the matrix order is even; the Chinese version says the same. This is a necessary consequence of its proposed existence characterization. Thus it suffices to exhibit a weighing matrix of odd order. We do not need to define its additional terminology concerning a window, a conversion constant, or a square decomposition.

A weighing matrix $W(n, k)$ is an $n \times n$ matrix with entries in $\{0, 1, -1\}$ such that

$$WW^T = kI_n.$$

This is the standard definition, for example in Harada and Munemasa [1]. The following witness has positive weight strictly less than its order.

Explicit counterexample. Let

$$W = \begin{pmatrix} 1 & -1 & -1 & 0 & -1 & 0 & 0 \\ 0 & 1 & -1 & -1 & 0 & -1 & 0 \\ 0 & 0 & 1 & -1 & -1 & 0 & -1 \\ -1 & 0 & 0 & 1 & -1 & -1 & 0 \\ 0 & -1 & 0 & 0 & 1 & -1 & -1 \\ -1 & 0 & -1 & 0 & 0 & 1 & -1 \\ -1 & -1 & 0 & -1 & 0 & 0 & 1 \end{pmatrix}.$$

All entries belong to $\{0, 1, -1\}$. Let $r = (1, -1, -1, 0, -1, 0, 0)$ and let ρ denote cyclic right shift on seven coordinates. Row i , for $0 \leq i \leq 6$, is $\rho^i r$. The seven cyclic correlations are

$$\begin{array}{c|cccccc} d & 0 & 1 & 2 & 3 & 4 & 5 & 6 \\ \hline r \cdot \rho^d r & 4 & -1 + 1 & -1 + 1 & -1 + 1 & 1 - 1 & -1 + 1 & -1 + 1 \end{array}.$$

Indeed, the nonzero contributions in each off-diagonal correlation are precisely the two terms displayed. A cyclic shift permutes the coordinates and preserves dot products, so

$$(\rho^i r) \cdot (\rho^j r) = r \cdot \rho^{j-i} r = \begin{cases} 4 & i = j, \\ 0 & i \neq j, \end{cases}$$

where exponents are taken modulo 7. Therefore $WW^T = 4I_7$ and W is a weighing matrix $W(7, 4)$.

Its order is $7 \equiv 1 \pmod{2}$, contradicting the conjecture's even-order requirement. Moreover,

$$0 < 4 < 7, \quad 4 + 2 = 6 \leq 7.$$

The witness consequently still disproves even order if one restricts to positive, nonfull weights and additionally imposes the conjectured lower bound $n \geq k + 2$.

Formal verification and semantic correspondence. The accompanying `Main.lean` uses Lean 4.19.0 and imports only `Std`. Its type `Matrix n` is `Fin n → Fin n → Int`, so it has exactly n rows and n columns. The definition `rowInnerProduct` sums $W_{it}W_{jt}$ over `List.finRange n`, which lists all column indices. Thus the second clause of `IsWeighingMatrix` is exactly the entrywise equality $WW^T = kI_n$; its first clause is the complete entry restriction. Integer entries and identities also give the same real matrix upon the canonical inclusion of integers into the reals.

The definition `W7` is the circulant specified above. The theorem `W7_displayed_rows` checks equality with every entry of the displayed array. The theorems `W7_entries` and `W7_orthogonality` check all 49 entries and all 49 row-pair inner products, respectively, using kernel-checked `decide`. They establish `W7_is_weighing`; `W7_odd_order` checks its odd order. The existential theorem `odd_order_counterexample` packages the actual matrix together with its positivity, nonfull weight, lower bound, orthogonality, and oddness.

The predicate `ClaimedEvenOrder` asserts that every weighing matrix has even order. The theorem `claimed_even_order_false` refutes it by the verified $W(7, 4)$; `even_order_claim_fails_under_bound` refutes even the restricted assertion just described. Finally, `full_conjecture_false` states that conjoining this false necessary assertion with any further assertions remains false. This is the logical bridge to the source statement, without replacing the matrix hypothesis by a mere numerical check.

Axiom audits are printed for all principal declarations. The primary matrix theorems depend only on the standard Lean axiom `propext`; the oddness calculation uses no axioms. There are no added axioms, admissions, or native decision procedures. The project is checked by a fresh `lake build` with warnings treated as errors.

Supplementary observation about the proposed lower bound. If the source’s undefined window lower bound is interpreted as requiring $n \geq k + 2$ for every weighing matrix $W(n, k)$, it also fails independently. The matrix

$$C = \begin{pmatrix} 0 & 1 & 1 & 1 \\ -1 & 0 & -1 & 1 \\ -1 & 1 & 0 & -1 \\ -1 & -1 & 1 & 0 \end{pmatrix}$$

has $CC^T = 3I_4$, zero diagonal, and off-diagonal entries in $\{1, -1\}$. It is a conference matrix and a weighing matrix $W(4, 3)$, whereas $4 < 3 + 2$. The Lean theorems `C4_is_weighing`, `C4_is_conference`, `claimed_necessary_bound_false`, and `conference_bound_false` verify this supplementary observation. This interpretation of the window plays no role in the primary odd-order counterexample.

References

- [1] M. Harada and A. Munemasa, *On the classification of weighing matrices and self-orthogonal codes*, arXiv:1011.5382, Introduction.